

Centralized Learning v.s. Federated Learning

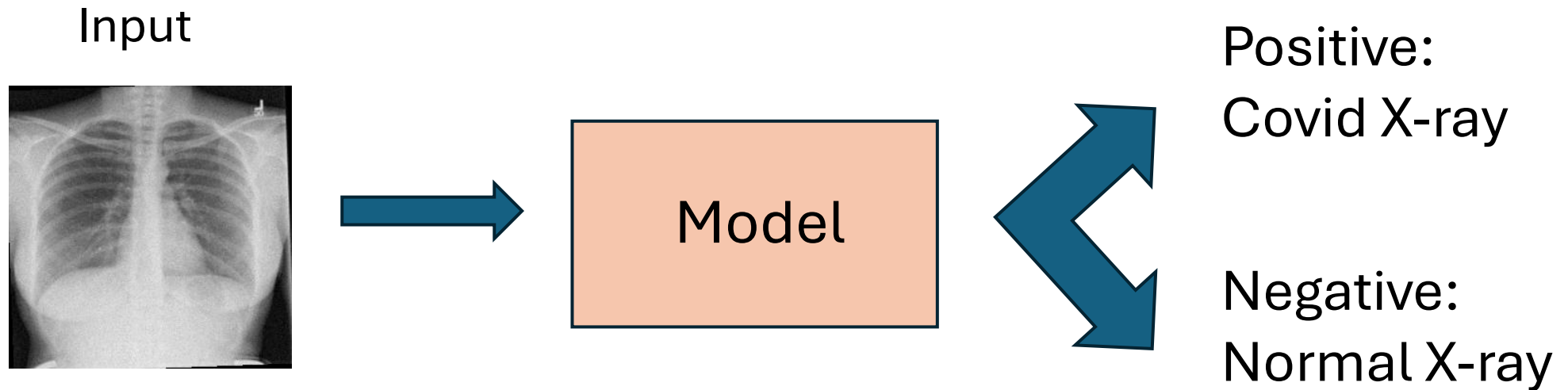
HW1

Due Date: 5/7 ~ 6/4 23:59

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HW Binary Classification



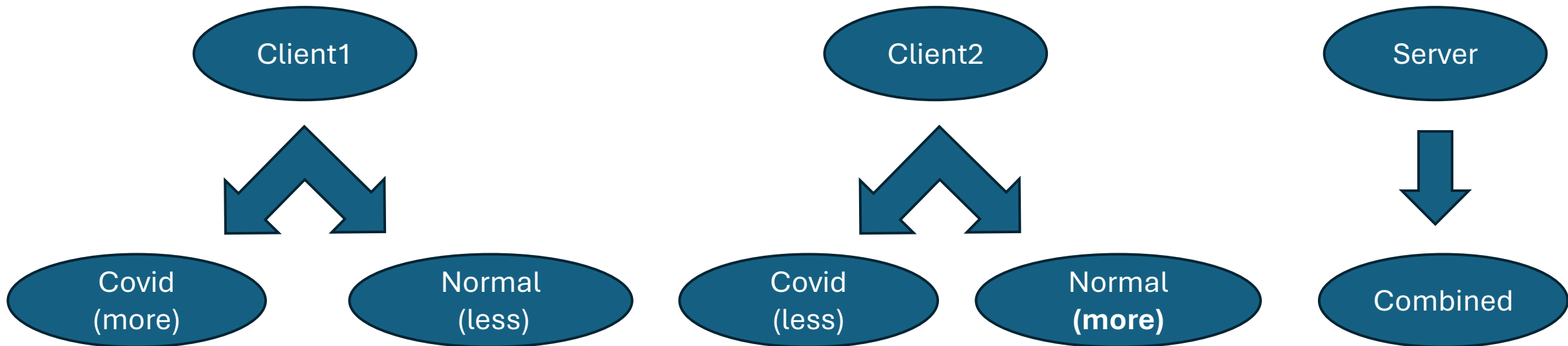
- Build the **Centralized Learning** and **Federated Learning** models
- Given 2 folders “CentralizedLearning”, “FederatedLearning”
 - Train the model and then predict the label of the input data
 - Use data in each folder for training each model. :
 - E.g., FL folder for FL model

Dataset Description

- A Preprocessed dataset (with random Gaussian Noise)
- In “CentralizedLearning” folder, we have task
 - COVID vs Normal
- Use the “train” set for training and the “test” set for evaluation.
- Please do not modify the training/testing data

Dataset Description

- In “FederatedLearning” folder, the data of the Client1 and 2
- The dataset is naturally imbalanced
- Please do not modify the training and testing data.



Evaluate Indicator

- Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- F1 score

$$F_1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}} = 2 \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} = \frac{2TP}{2TP + FP + FN}$$

- Confusion Matrix

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

Implement Details – Centralized Learning

- Fill in the function blocks which have “TODO” inside.
- Implement
 - Model architecture
 - The “train” function and cross validation
- Calculate the F1 score, plot the confusion matrix
- You should show us **the results**:

Accuracy, loss, F1 score of task

P.S. You don't need to print all three results at once during a single training session; you can print them separately.

Implement Details – Federated Learning

- Fill in the function blocks which have “TODO” inside.
- Implement
 - Model architecture
 - weight_aggregation
 - round_robin_training and cross validation
- Use Round Robin as the exchange protocol
- Calculate the F1 score, plot the confusion matrix
- You should show the accuracy, loss and F1 score of the Client1, Client2 and Server

TODO Example

```
class Model(nn.Module):
    # TODO: Implement your own model
    def __init__(self, num_classes=2):
        super(Model, self).__init__()

    def forward(self, x):
        return
```

```
def weight_aggregation(self, client_weights):
    # TODO: Implement the weight aggregation
    return

def round_robin_training(self):
    # TODO: Implement the round-robin training process and cross validation
    self.plot_loss()
    return
```


Report

- You should show your **Accuracy**, **Loss** and **F1 score** in your report. Screenshots or charts are acceptable.
- If you modified the original code, please tell me what you have changed in your code.
- **Please add a description about how to run your code.**
 - E.g.: requirements & python version .etc
- Please write the report **clearly**; otherwise, I will deduct points as necessary.



```
requirements.txt X 11452
requirements.txt
1 torch>=2.0
2 torchvision>=0.15
3 scikit-learn>=1.2
4 matplotlib>=3.7
5 seaborn>=0.12
6 tqdm>=4.65
```

Scoring Criteria

- . Report: 30%
- . Private test results: 40%
- . Performance: 30%

Conclusion

- Please hand in your **code**(py files) and **report**(pdf) to eeclass.
- Please rename your files.
 - Ex: Student ID: 114522097 -> 114522097_CL.py
114522097_FL.py
114522097_report.pdf
- **Late submissions are not allowed.**